FARES SHRETAH

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PROFILE

MSc Machine Learning student at UCL (graduating September 2027) with research experience at the Max Planck Institute and industry experience building LLM agents. Focus on fine-tuning and evaluating language models and on agentic systems.

EDUCATION

University College London (UCL) **Sep 2026 – Sep 2027 (expected)**
MSc Machine Learning · ~~Said~~ Saïd Foundation Scholar

Université Paris-Saclay **2023 – 2026**
BSc Computer Science · Grade: 15.43/20, top 4 in cohort
Coursework: Machine Learning, Data Science, Advanced Algorithms, Symbolic AI, Fairness in AI

EXPERIENCE

Machine Learning Research Intern, Max Planck Institute for Human Development **Jun 2026 – Aug 2026**
Berlin, Germany · Full-time · PyTorch, Hugging Face Transformers, vLLM, SLURM, Python

- Led a self-directed ~~research~~ project on conversational mode collapse in LLMs~~(models in a closed loop degrading into repetitive text), testing whether a model can detect the collapse state~~ : can a model detect when a closed-loop conversation has become repetitive, and recover from it~~.~~ ?
- ~~Designed the methodology and timeline,built a~~ Built an 8,504-example dataset from 10,~~000-example training dataset, and~~ 000 self-conversations, labelled by a 31B LLM judge (95% validated), and fully fine-tuned ~~Qwen 2.5 3B on the institute's HPC cluster using SLURM.~~
- ~~Extended an existing NLP research codebase with an LLM-as-a-judge evaluation and ran the models with vLLM.~~ Qwen2.5-3B over five versions on the HPC cluster (SLURM, vLLM).
- Result: the model detects ~~its own~~ collapse in unseen conversations~~and switches to a new topic~~, was preferred over the previous best in 11 of 15 paired runs, and cut empty turns from 2.8% to 0.4%; 500-turn runs showed recovery is temporary.

Agentic Engineer, To Ai **May 2024 – Oct 2025**
Beirut, Lebanon · Part-time · LangChain, LangGraph, Python, AWS, RabbitMQ, Meta APIs

- Built LLM agents for booking, sales and customer support with LangChain and LangGraph, using dynamic routing and prompt orchestration~~.~~
- ~~Wrote~~ ; wrote the prompts and ~~created the example data used to guide agent behavior~~example data that guide agent behaviour.
- Built a message-queue architecture on AWS SQS and RabbitMQ to process webhook events reliably.
- Led development of the company's production website (to-ai.ai).

Co-Founder, Velora-Dev **Aug 2025 – Present**
Freelance web and AI studio, Lebanon (Remote) · Next.js, TypeScript, Supabase, SQL

- Built CRM dashboards, AI-based company workspaces and AI agents for client companies.
- Delivered full-stack apps, including Silver Rent (car rental platform with booking logic, payments and email confirmations) and the official portal of Hasroun Municipality.

PROJECTS

Image Captioning with Transformers (GitHub) **PyTorch**

- Built a multimodal captioning model: pre-trained ResNet-50 encoder with a custom Transformer decoder using cross-attention.
- Implemented the full pipeline: preprocessing, attention-based decoding, multi-reference BLEU evaluation, and a Flask app that ~~visualizes~~ visualises attention heatmaps.

ATP Tennis Match Prediction (GitHub) **Scikit-learn, Pandas**

- Built an ML pipeline with feature engineering on historical ATP data to predict match outcomes.
- ~~Built CRM dashboards, AI-based company workspaces and AI agents for client companies.~~
- ~~Delivered full-stack apps, including Silver Rent (car rental platform with booking logic, payments and email confirmations) and the official portal of Hasroun Municipality.~~

SKILLS

- **ML / LLMs:** PyTorch, Hugging Face Transformers, fine-tuning, LLM-as-a-judge evaluation, vLLM, LangChain, LangGraph, Scikit-learn, Pandas, NumPy
- **Languages:** Python, C, C++, Java, OCaml, JavaScript / TypeScript
- **Infrastructure & Web:** SLURM (HPC), AWS, Docker, Git, RabbitMQ, SQL, REST APIs, Next.js

~~HONORS~~ HONOURS & LANGUAGES

- Merit Scholarship (Excellence); Top 10 student, two consecutive years; Academic Tutor
- **Languages:** English (fluent, IELTS 8.0), French (fluent), Arabic (fluent)